Rehabilitation Exercise Game Model for Post-stroke using Microsoft Kinect Camera

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***Abstract*— The purpose of rehabilitation is to improve the movement of stroke patients so that they can independently perform daily tasks on their own. Conventional therapy is a tedious exercise which usually reduces the motivation of the patient to do the exercises. Video game is a way that can motivate and help patients recover their motor skills, and also act as a deterrent pain. Video games have been used in physiotherapy, occupational therapy, and psychotherapy. In order to increase mental satisfaction and physical vitality in rehabilitation therapy, some therapists have tried to use the off-the-shelf digital game devices in rehabilitation and have found effective treatment outcomes in addition to enhancing patients' treatment motivation. Although many studies have been done in this field, there is still lacking the existing recovery game, because too many different demands must be met for each type of therapy. There is one disadvantage at the beginning of the game because patients have difficulty understanding how to use it. The current recovery games are less fun and participative for the user, while the popular games have fewer components to use in rehabilitation training. Furthermore, post-stroke patients find it difficult to get a friend to play or spend time with them every day, which will make them feel isolated. Therefore, rehabilitation exercise games that support multi-player will provide a higher motivation than the single-player. Since most stroke patients suffer from cognitive impairment, cognitive challenge levels are also the key factors in the design of the game so that it does not become an obstacle for the recovery process. Therefore, this study develops a prototype of a rehabilitation exercise game that contains aspects of the social context, the type of movement and cognitive challenges. It also provides usability in game design, according to a post-stroke stage so that they can perform recovery activities based on their ability. In addition, this study highlights technology and rehabilitation exercise games in Malaysia.**

***Index Terms*—rehabilitation exercise games, Kinect, post-stroke**

I. Introduction

Stroke is the main reason that leads to disability for adults over a long period. About 80% of post-stroke patients have disability of nerve movement [1]. Post-stroke patients, who have disability in nerve movement, will have difficulty in controlling the movement, strength, and range of motions. This weakness prevents them from doing daily tasks on their own, e.g. wearing their clothes, combing their hair, and taking a shower. To overcome this problem, continuous rehabilitation is needed so that it can strengthen their nerve movement. By doing the exercises continuously, it can help the patients to recover, and become independent to do their daily tasks. Exercises can be used to improve motor movement and strengthen body movement. The emphasis on task specific training is important because research has shown that stimulating practical movements with goals achieves better success than through repetition alone [2]. However, patients always have less motivation to do the exercises that have been scheduled.

Recently, computer games and simulations have been recognized as a motivational tool in the recovery process. However, a serious game should meet the needs and requirements so that it can be used in the recovery process. A large number of the most critical needs, which are not available in commercial games, can be adjusted in the systems or game setup and configuration, e.g. calibration on the input and speed of the game [3]. This means that the various aspects of the game can be selected, e.g. a suitable challenge, motion analysis, patients' report and others. Therefore, the display configuration should be developed so that users can apply visual setting without the need for programming or design skills.

By using the technology, it can motivate patients to find pleasure, make motor control moves repeatedly, and shift the focus of the patient from feeling pain during exercise. To this end, the researcher created a system recovery using the Kinect sensor to enhance the exercises using natural exposure, which allows the body to interact with the free movement [4]. Kinect can be defined as software that makes it possible to capture, track, and interpret body movements, gestures, and voice [5]. In other words, using Kinect, patients do not need the keyboard, mouse, and video game controller that are burdensome and would restrict their movements.

II. Objectives

The main objectives of this research are: (1) to identify conventional rehabilitation exercise for post-stroke patients, (2) to design rehabilitation games that match conventional rehabilitation exercise, and (3) to test the effectiveness of rehabilitation games. The findings of this research have some significance for the stroke rehabilitation program in PPUKM. They are: (1) the post-stroke patients will be more motivated as

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regards the rehabilitation exercises using Kinect, while the rehabilitation game was designed to suit patients; (2) the number of workers who monitor the rehabilitation activities will be reduced. The scope of this research incorporates: (1) the design of the prototype game involving post-stroke patients at Universiti Kebangsaan Malaysia Medical Centre (PPUKM) as samples – the game can be used by post-stroke patients who experience the same problem; (2) this prototype only focuses on certain areas only, which is the upper limb.

III. Literature Review

In Healthcare there are many technologies used in stroke recovery programs. The conventional technology which identifies body language only focuses on the bookmark system; it recognizes the shape of human hands and controls the cursor. However, based on studies that have been made, they focus mostly on problems during the recovery program; the patient cannot move freely because he or she is bound by the tools during the recovery program, which are also a burden.

The Multi-Flexor is one of the technologies that have been used for rehabilitation exercise. The tool uses the motor as a generator and gearbox together to provide an adequate round of movement. The design of this tool can be adjusted with the position of the links or just installed inside or outside of the different modules. The patient can do all the exercises for the wrist and the elbow joint [6]. The problem of this technology is that the design of mechatronic devices does not reach a level where the device can be used by patients to exercise every day. Another technology that has been used for rehabilitation exercise is the Nintendo Wii. This appropriate control device is designed for patients who suffer from foot problems, requiring ability in the problematic foot while controlling elements in the game. It means physical functions necessary to replace the remote that controls the game [7]. The problem of this technology concerns the physical movement required of the controller as determined by the game designers. Indirectly, it becomes a burden to the patient. Figure 1 and Figure 2 show the examples of Nintendo Wii games.



Figure 1. Nintendo Wii Games - Wii Sports



Figure 2. Nintendo Wii U Games

The virtual hand and 3D models are further technologies used for rehabilitation exercise. This mobile device consists of a 3D camera and sensor worn on the body where it can help to control body movements during recovery training through the display [8]. The problem of this technology is that the patients are bound by the device and cannot do free movement according to their capabilities. In addition, technologies used for rehabilitation exercise include Wii Remote. This technology detects the patient's body and using algorithm detection reads the acceleration of the Wii Remote, which is in the patient's arm and body [1]. The problem with this technology is its limitation because the controller is mounted on the body of the patient. The solution for this problem is to use Kinect technology, which can detect the position of the patient's body without the need for any control device or installing the device on the patients' body.

There are many examples of games that use virtual movement to control the game. They are:

- Balloon Pop: Patients need to put their hand on the balloon for a few seconds until it bursts. When all the balloons are burst, the patients can go to the next level where they only need to burst a balloon of a particular color [9].
- Path Follower: This game can improve balance while walking. Patients should follow established routes. If successful, they need to go along the road and the difficulty of the road is increased [9].
- Duck Duck Punch: Patients reach for objects on the screen using a virtual hand. This game uses the Kinect technology [10].
- JewelMine: Patients reach for a flaming diamond. Points are accumulated and the performance recorded for training for the next restoration [11].
- Mini-Games Temple of Gupta, Face of Chronos, and Three Wind Gods: These are three mini games. First, the player runs through the temple to collect artifacts and avoid the obstacles. Second, the player needs to

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climb a mountain with arms up above to collect artifacts. Finally, the player mimics the character by doing head movements [12].

However, the game only focuses on the recovery movement training; there is no element of fun as in other commercial games. In addition, the most existing recovery game only involves one player at a time. Also, there are factors which cause the patient to be less motivated to perform the recovery, where there is no social context. The great influence in the social context in stroke rehabilitation is the design of the game itself, although it is a complicated process. The designers should not only understand the motivation of individuals, but the game should also be designed to motivate the physical needs and social context in a particular technology. Patients are only bound by these tools as their own motivation, because of the social relationships and involvement of family members [13]. Figure 3 and Figure 4 show the examples of Kinect games.



Figure 3. Kinectimals



Figure 4. Kinect Sports

IV. Methodology

To gather data and obtain the right information, the following is a draft study conducted during the research process from the beginning to the end.

A. Analysis

Analysis phase is the basis of the game design. At this stage, the research problem and research objectives are identified. In this phase, the research data are collected by interviewing the patients, understanding their interests and the technical aspects of their stroke rehabilitation exercises, as well as talking to a therapist. In addition, the observations are conducted in the stroke rehabilitation center to get a clear picture of how the stroke rehabilitation program is run. In the context of the construction of the game design, there are some basic questions in the analysis phase:

- Who is the target user of a rehabilitation exercise game?
- Which game design is suitable for the patients?
- What kind of problems or limitations need to be overcome?
- What results are expected?

B. Design

In this phase, the game design is identified based on data and information obtained during the analysis phase. A discussion with the therapist is essential so that the appropriate games designed match the rehabilitation exercises. Elements of the social context, types of games, cognitive challenges, storyline and NPC are combined.

C. Implementation

In this phase, the game design that has been identified is ready and the rehabilitation exercise developed.

D. Testing

In this phase, the game is tested with the involvement of selected post-stroke patients, and the observation and analysis of the effectiveness conducted by a therapist. In addition, feedback from post-stroke patients who are involved is analyzed. Finally, the conclusion is based on data that has been analyzed.

Figure 5 below shows a summary of how the research is conducted and how the methods map and fulfill the objectives.

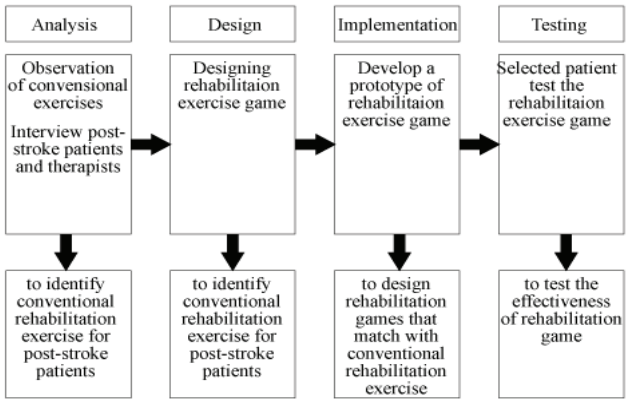


Figure 5. Summary of research plan

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V. GAME DESIGN AND PROTOTYPE

Motivation is key in stroke rehabilitation [13]. To motivate patients to perform rehabilitation exercises, games should have an interesting element. These are a few of them: (1) have a target that allows the patient to perform an extreme movement, (2) ensure the challenging game is simple but has a difficult task, (3) provide interest and appropriate feedback during the game, (4) provide punishment and reward [1].

Social and emotional support from family members, relationship with therapists, changing responsibilities in the family, understanding during the recovery process, whether in the short term or long term, and music are important motivation in the recovery process and can speed up the process of recovery from stroke [14]. To make the rehabilitation exercise game more interesting and the patients more motivated to do rehabilitation exercises, here are some criteria [15]:

- The sound and the display are important: In the results of the study, he found that it is important to focus on the issue of size and color. A patient tried to focus on recovery while playing the game. However, when he played a recovery game with nice images and sounds, automatically he could give full attention to the game.
- The level of difficulty can be adjusted according to the patient's level of ability: For patients who are familiar with the game of recovery, they will easily get bored unless the difficulty can be raised manually. Therefore, there should be automatic setting where the level of difficulty will vary according to the success or failure of the patient in the game based on data provided by the therapist.
- Non-Player Character (NPC) and the storyline to make the game more interesting: Researchers believe that the opportunity to interact with the NPC, and create a more dynamic storyline, can give a sense of depth to the patients so that they can immerse themselves in the game.

Meaningful play, challenge and interaction technology have been identified as four attributes of the game design principles. These principles are found important for brain damage rehabilitation and have been used in developing a kinect games prototype for cognitive rehabilitation of acquired brain injury, specifically attention, concentration, executive functions, eye-hand coordination and memory deficiencies[17].

A. Meaningful play

Meaningful play emerges from the relationship between player action and game system's outcome. This relationship should be discernible and integrated, so that a player may notice how his selection is affecting the game from the start to the final stages and this can be achieved through feedback. Feedback is how the game responds to the player's actions and is important in the creation and maintenance of meaningful play. In addition, it enables the player to measure his skills' progress. Feedback is categorized into three; aural, visual and haptic which present correct or incorrect choices or responses. Elements like progress bars, numerical scores, sound and character dialogue are all considered as feedback that may increase player's motivation, enjoyment and generating a desire to complete particular tasks to attain a certain goal.`

In addition, the user can be offered a good mental model by considering four properties: visibility, affordance, mapping and feedback. These principles have gained importance for user interface design and are actually used by many researchers for designing virtual environments for brain injury cognitive rehabilitation.

B. Challenge

The game challenge should match the player's abilities as well as his skills. The cognitive load theory has considerable implications for the design of rehabilitation intervention for people with ABI. New rehabilitation intervention should cause as little cognitive load as possible so that the patient can concentrate completely on the task. Hence, game based intervention should be customized to begin at a stimulus challenge level that is attainable and comfortable for individuals, and then proceed with a gradual progression of challenge that is informed by their performance.

On other words, at the start of a new game, the player first pursues a low level of challenge as he is not yet familiar with the game. Every time, the player plays the game, his familiarity and skills increase and then in order to keep on enjoying, he/she asks for a more challenging level. A difficult level may cause the player to frustrate or even quit the game. On the other hand, players feel no interest in a boring or low leveled challenge.

In short, as shown in Figure 6, the level of challenge should be designed in such a way that it is ideal for the individual player to compete with the challenge which would have been increasing at a steady rate [18]. In addition, in accordance with flow theory, being fully engaged is derived from activities that are optimally challenging where an individual's skills are coordinated with the challenges faced with clear goals and feedback.

Most games deal with this problem by incorporating levels to control the difficulty. In order to advance to the next level, specific skills along with an understanding of the game mechanics plays a crucial rule. In addition, some games modify the challenge according to the patient's performance. In rehabilitation, the therapist may modify the game challenge to suit the patient's abilities judging from his progress and interaction [19].

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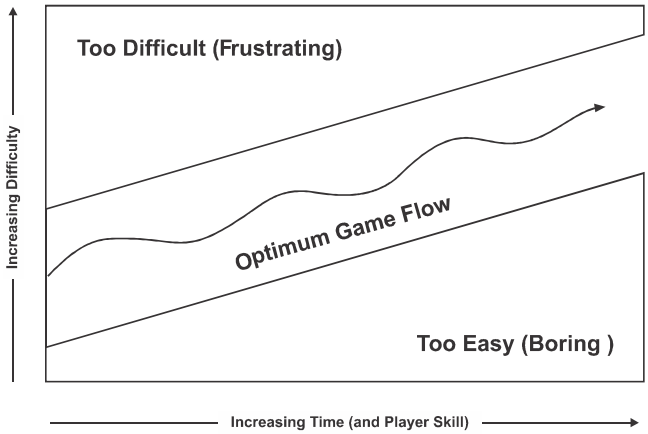


Figure 6. Game Flow

C. Interaction Technology (Kinect Gesture)

Another fundamental characteristic is interaction technology which is the main criterion for the classification of serious games in rehabilitation area. It is actually the technology for patient to interact with the system. Natural user interfaces are the most suitable interfaces to be employed in rehabilitation as patients' various cognitive and motor disabilities result in their difficulties in interacting with the system.

D. Game Prototype

To sum up, these three game design principles (meaningful play, challenge and interaction technology) should be taken into consideration for making kinect games effective in the process of rehabilitation.

Figure 7 shows the game prototype. Here, in this mini game, player has to follow a sequence of sounds and colors produced. The aim of this mini game is to improve memory, attention and concentration skills. Furthermore, it is played by clicking on a colored box using hands in the same sequence they were first played.



Figure 7. Kinect Coloured Box Game

Figure 8 shows the puzzle game prototype for improving concentration and problem solving skills. It is played by organizing small images in the correct order to have a full picture in a few possible steps. As the game begins, small images are scattered in random order on the grid. The player needs to click on one image to activate it. He then clicks on the desired location and continues to do so until all images are placed in the desired locations.

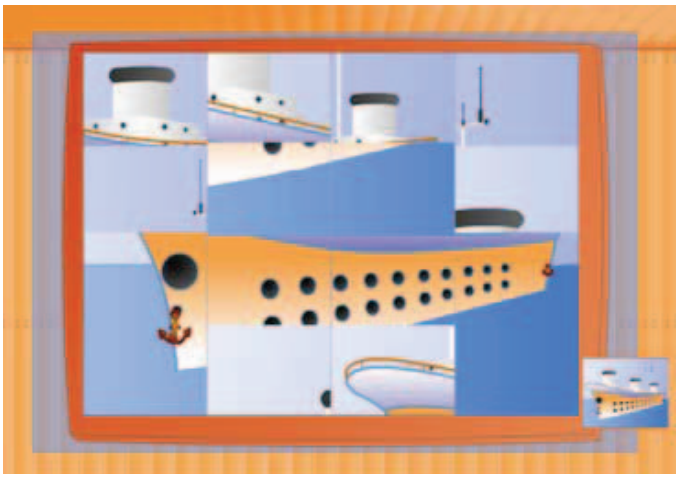


Figure 8. Kinect Puzzle Box Game

VI. Conclusions

According to researchers, constant exercise movements can provide a stimulus to the brain and repair motor control. However, the number of exercises performed in the medical center is limited. To overcome these problems, the therapist will always recommend home exercise therapy as an alternative exercise. For patients who do not exercise at home as recommended, the recovery will be slow and they will be less motivated. Therefore, a video game that uses movement to interact can make patients more motivated; it is one of the alternative ways to do additional rehabilitation exercise. Many researchers agree that training involves elements of game in stroke rehabilitation, and the patient will be more motivated to do the rehabilitation exercise [4][16].

This study developed a prototype rehabilitation kinect games that contains aspects of social context, type of movement, and cognitive challenges. It will also provide a suitable game with a post-stroke stage so that they can perform recovery activities based on their ability. The effectiveness of this rehabilitation exercise game will be determined after obtaining feedback from patients and therapists who have been involved in the study. Game-based rehabilitation interventions are most effective when they are shaped to meet a patient's particular needs and preferences, not when they are prescribed as an isolated intervention of addressing particular cognitive and/or physical deficiencies. This process needs to begin with a comprehensive evaluation that gives the rehabilitation professionals an opportunity to assess the patient's ability, needs, preferences and

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expectations. The evaluation generates information that can assist in tailoring game interventions which further enhance the quality of training and positively affect the self-esteem and motivation of patients.

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